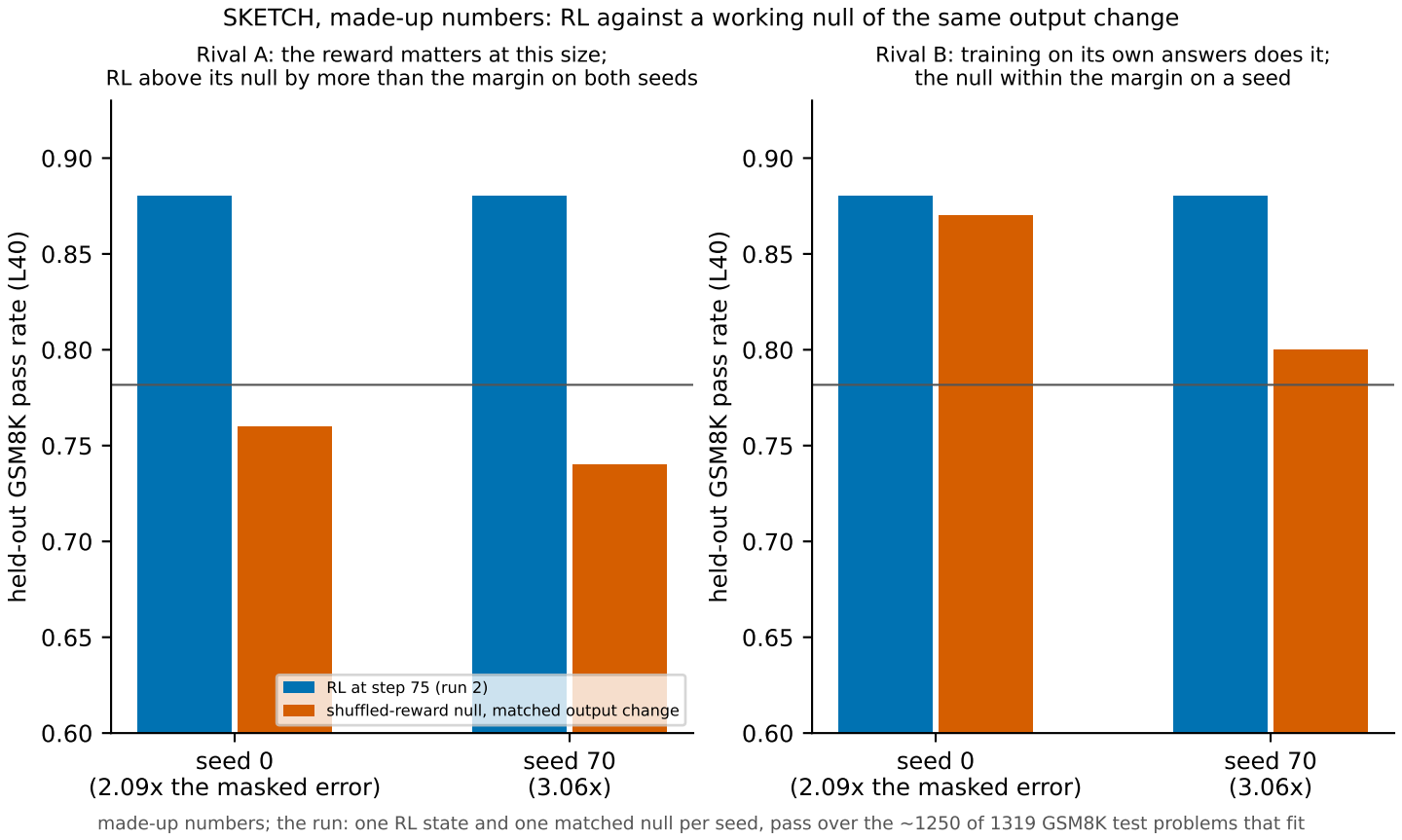
SKETCH, made-up numbers: RL against a working null of the same output change
Rival A: the reward matters at this size;
RL above its null by more than the margin on both seeds
Rival B: training on its own answers does it;
the null within the margin on a seed
held-out GSM8K pass rate (L40)
0.90
0.85
0.80
0.75
0.70
0.65
0.60
RL at step 75 (run 2)
shuffled-reward null, matched output change
seed 0
(2.09x the masked error)
seed 70
(3.06x)
made-up numbers; the run: one RL state and one matched null per seed, pass over the ~1250 of 1319 GSM8K test problems that fit